

## Article

# Macro–Market Fusion with Cross-Attention for Equity Return Prediction

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## Abstract

Macroeconomic conditions are widely believed to influence the direction of equity markets, yet most forecasting models either ignore macroeconomic information or incorporate it through a small set of ad hoc predictors. We propose XAttnFusion, a macro–market fusion architecture that jointly learns from high-frequency market data and lower-frequency macroeconomic time series for equity return prediction. The model comprises three branches: a 1D convolutional network that encodes 40-day market windows (price, volume, and technical indicators), a temporal convolutional network that encodes 24-month macro sequences, and a feedforward branch for volume-at-price structure features. These representations are integrated through multi-head cross-attention, in which the current market state queries the macro sequence to produce a fused representation for directional forecasting. We evaluate XAttnFusion on daily SPY returns from 2012 to 2024 using purged cross-validation with a 5-day embargo to prevent information leakage. To address potential look-ahead bias from macroeconomic publication lags, all macro inputs are lagged by two months. The model achieves a mean out-of-sample AUROC of  $0.63 \pm 0.05$ , representing a 27% improvement over random and an 8.1% improvement over the best concatenation baseline. In a fair comparison where each model is independently hyperparameter-tuned, cross-attention fusion improves AUROC by 0.047 over concatenation ( $p = 0.031$ , Wilcoxon signed-rank test). The model also generalizes to QQQ and IWM, where cross-attention consistently outperforms concatenation fusion. Crucially, the model’s discriminative ability is state-dependent, indicating that the value of macro–market fusion is itself conditioned on market structure. Permutation-based feature importance shows that macro and market branches contribute on a comparable scale (approximately 48% and 36%, respectively), so the gains come from jointly fusing two comparably weighted sources rather than from a single dominant input. Our results show that explicitly modeling macro–market interactions with interpretable attention improves predictive accuracy over naive fusion strategies and provides insight into the time-varying relevance of macroeconomic information in financial forecasting and equity market prediction.



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**Keywords:** macroeconomic variables; equity return prediction; financial forecasting; deep learning; cross-attention; time-series forecasting; interpretability; market structure; volume-at-price (VAP)

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## 1. Introduction

Financial markets are shaped not only by asset-specific information and order flow but also by the broader macroeconomic environment. Variables such as inflation, output growth, monetary policy, and risk sentiment are widely believed to drive time variation in expected returns and risk premia [1,2]. Yet, despite this consensus in practice, building models that systematically incorporate macroeconomic information into high-frequency or asset-level return forecasts remains challenging. Macroeconomic time series are inherently noisy, are observed at lower frequencies than market data, and often display complex long-range temporal dependence. In practice, these properties make macroeconomic information difficult to use effectively in empirical models. Many empirical studies in asset pricing and forecasting therefore either omit macroeconomic variables altogether or compress them into a small set of factor-style proxies with hand-chosen lag structures.

Recent work in deep learning for time-series forecasting shows that neural networks can recover nonlinear patterns in price histories and related market features [3–5]. Financial applications now use convolutional neural networks (CNNs), recurrent architectures such as LSTMs [6], temporal convolutional networks (TCNs) [5], and transformer-based attention models [7]. Even so, most of this literature still works with price-only or market-only inputs. When macroeconomic variables appear, they are usually stacked alongside market features and passed through a single shared network. That setup gives the model more inputs, but it does not force it to learn a separate macroeconomic history or a clear representation of how macro conditions alter market signals over time [4,8]. It also makes interpretation harder. In most cases, the model still behaves like a black box, and it is difficult to tell which macro variables or which parts of the historical window are driving the forecast [9,10].

In this paper, we propose a multi-modal neural architecture that is built around that gap. The model uses two specialized branches. A CNN branch learns local temporal patterns in market data, while a TCN branch encodes longer-horizon dynamics in macroeconomic time series. We then connect the two through a cross-attention mechanism in which the market representation queries the macro representation. This choice matters for more than prediction accuracy. It lets the model learn, from the data, which macro variables and which historical lags deserve weight for a given forecast, and it yields an attention pattern over macro history that can be inspected directly.

The framework is flexible enough to be used with different asset classes, forecast horizons, and macroeconomic datasets. Alongside standard price-based inputs, we include structure-aware features such as volume-at-price (VAP) measures that reflect aspects of the market microstructure. We evaluate the model with purged cross-validation and a temporal embargo. This design reduces look-ahead bias and gives a stricter view of out-of-sample performance.

The main contributions of this paper are as follows:

- We introduce a cross-attention mechanism for macro–market fusion that captures how macroeconomic conditions shape market-based signals, rather than applying deep architectures to price histories alone.
- We develop a multi-modal architecture consisting of a CNN branch for market data, a TCN branch for macroeconomic data, and an attention-based fusion module, which allows the model to use complementary inductive biases across data sources.
- We provide interpretable attention over macro lags, making it possible to identify which macro variables and historical periods receive the most attention weight in current predictions, and offering a diagnostic lens on the temporal structure of the model’s macro usage.

- We incorporate structure-aware features, including volume-at-price (VAP) measures, and evaluate the model using purged cross-validation with a 5-day embargo, yielding more credible estimates of out-of-sample performance than standard random splits.
- We show empirically that predictive performance is state-dependent: the model is most discriminative when price is near high-volume support nodes and least effective in neutral regions, indicating that the value of macro–market fusion depends on the underlying market structure.

The remainder of this paper is organized as follows. Section 2 reviews the related literature on macro-based return prediction, deep learning for asset pricing, multi-modal and attention-based time-series models, and interpretability in financial machine learning. Section 3 formalizes the prediction problem and describes the proposed macro–market fusion architecture in detail. Section 4 presents the data, feature construction, and experimental design. Section 5 reports empirical results and baseline comparisons. Section 6 discusses the macro-level insights obtained from the attention structure. Section 7 presents ablation analyses isolating the contribution of each model component. Section 8 concludes and outlines directions for future research.

## 2. Related Literature

Our work relates to several strands of the literature in empirical asset pricing, macro–finance, and machine learning for financial time series. We organize this section around four themes: (i) return prediction with macro variables and factors, (ii) deep learning for asset pricing and return forecasting, (iii) multi-modal and attention-based architectures for time series (including financial applications), and (iv) interpretability in financial machine learning.

### 2.1. Return Prediction with Macro Variables and Factors

Asset pricing research studies whether macroeconomic variables and risk factors can explain and predict returns. In the arbitrage pricing tradition, macroeconomic state variables such as industrial production, inflation, and term premia are viewed as sources of systematic risk [11]. Factor models, including the Fama–French three-factor model [12] and its extensions, combine firm characteristics with common risk factors to explain cross-sectional variation in expected returns.

Macro–finance focuses on how macroeconomic conditions drive time variation in expected returns and risk premia. Consumption-based and habit-based models link this variation to aggregate consumption risk and changes in risk aversion [1]. Empirically, a common approach is to extract a small number of latent macro factors from large data panels and use them for forecasting and asset-pricing tests [13,14]. Surveys of return predictability emphasize the importance of out-of-sample evaluation and document that predictive relationships are often unstable across periods [15,16]. Consistent with this concern, Ref. [17] shows that many predictors weaken once evaluated out of sample, while international evidence suggests that predictability varies across markets [2]. We view this evidence as highlighting instability rather than the absence of macro information.

Data timing introduces an additional complication. Many macroeconomic series are released with delays and are subject to revisions, so the real-time information set differs from ex post data. Forecasts based on revised data can therefore overstate predictability. Using real-time macro releases, Ref. [18] finds evidence of predictability once nonlinearities are incorporated. Related work shows that macro information is concentrated around announcement periods [19,20].

Our approach builds on this literature but differs in how macro information enters the model. Instead of selecting a small set of predictors or latent factors, we input a high-

dimensional panel of macro variables and allow the model to learn both representations and interactions with market data endogenously. This shifts the design from ex ante variable selection to data-driven representation learning.

## 2.2. Deep Learning for Asset Pricing and Return Forecasting

Machine learning models are now widely used in asset pricing and return prediction. Ref. [3] shows that flexible learners outperform linear models when interactions across characteristics are important, and Ref. [21] uses deep networks to learn latent factors. Ref. [4] further incorporates a no-arbitrage structure into deep learning frameworks.

On the time-series side, neural networks have been studied as forecasting tools, with mixed evidence on when they provide gains [22,23]. Early work uses recurrent models such as LSTMs [24,25], while more recent studies explore transformer-based architectures [8]. A separate literature examines high-frequency prediction using order book data [26,27], where recent work emphasizes that improvements in statistical metrics do not necessarily translate into economic gains [28]. Related research also highlights the importance of uncertainty quantification in return forecasts [29]. Surveys emphasize tradeoffs between flexibility, interpretability, and robustness [30,31].

Most of this literature treats macro variables, when included, as additional features within a single representation. Our framework differs by explicitly separating macro and market inputs and learning their interaction through a dedicated fusion mechanism. This design forces the model to construct a distinct macro representation rather than embedding macro information implicitly within a large feature set.

## 2.3. Multi-Modal and Attention-Based Time-Series Models

Attention mechanisms and transformer architectures [7] have become standard tools in sequence modeling. A large literature develops variants tailored to time series, including locality-aware transformers [32], efficient long-sequence models [33], decomposition-based architectures [34], and interpretable multi-horizon models [35]. Later work introduces frequency-domain methods [36], patch-based tokenization [37], and cross-variable attention designs [38,39]. Empirical comparisons suggest that performance depends heavily on architectural choices rather than any single dominant design [40].

Multi-modal architectures extend these ideas by combining heterogeneous data sources. A common design encodes each modality separately and fuses them using attention. In finance, attention-based models have combined price data with text, order flow, or graph structures [41–44]. However, macroeconomic time series are typically included as auxiliary covariates rather than as a fully independent temporal modality.

Our contribution is closest to this multi-modal attention literature but differs in a key dimension: we treat macroeconomic data as a separate temporal stream and explicitly model macro–market interaction through cross-attention. This design places structural pressure on the model to learn how macro history influences market dynamics, rather than allowing this interaction to remain implicit.

## 2.4. Interpretability in Financial Machine Learning

Interpretability has become increasingly important in financial machine learning. Surveys document a growing use of post hoc explanation tools such as LIME [45] and SHAP [46], as well as model-based approaches [47–49]. In asset pricing, interpretability is tied to economic plausibility, as predictions should align with reasonable mechanisms [9]. Some recent work embeds interpretability directly into model design [50].

Deep learning studies often assess importance through perturbation or factor analysis [3,4], but these approaches typically lack a time-indexed interpretation of inputs.

Attention mechanisms provide such a structure, though their interpretation remains debated [10,51].

Our framework contributes by generating attention over macro lags, providing a time-structured measure of how macro history enters predictions. Importantly, we interpret these signals cautiously as model-based importance measures rather than causal effects.

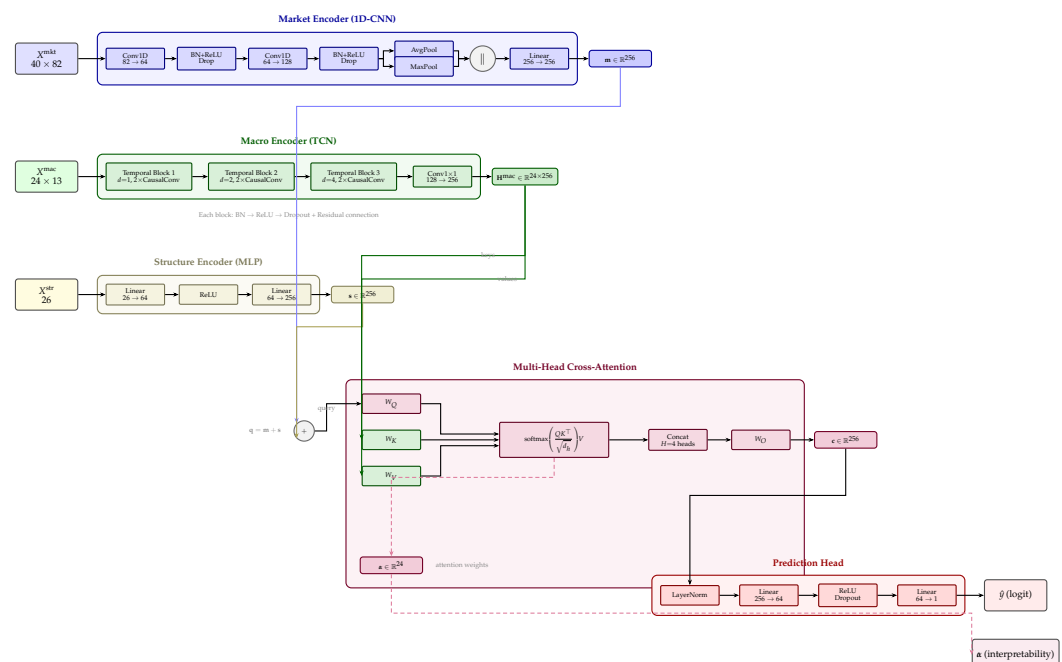
### 2.5. Our Contribution Relative to the Literature

The literature suggests that macro variables contain useful but unstable information, that deep learning improves performance in high-dimensional settings, and that attention mechanisms provide flexible sequence representations. Existing financial applications typically either focus on price-based features or incorporate macro variables as additional inputs within a unified feature space [3,4,8]. Multi-modal attention models exist, but they more commonly combine price data with text or graph information rather than treating macroeconomic series as a primary modality [41,42].

We build on these strands but differ in three ways. First, we maintain separate representations for macro and market data. Second, we use cross-attention to model their interaction explicitly. Third, we combine this architecture with a time-series cross-validation framework that respects the information set available at each point in time. This design allows us to quantify the incremental contribution of macro information while preserving interpretability and the temporal structure.

## 3. Methodology

In this section, we formalize the prediction problem and describe the proposed macro-market fusion architecture (Figure 1). We first introduce notation and the overall problem setup, then detail the market branch, the macro branch, the structure branch, and the cross-attention fusion mechanism, followed by the prediction head and training objective.



**Figure 1.** XAttnFusion architecture. Three input branches encode market features (1D-CNN), macroeconomic sequences (TCN), and volume-at-price structure (MLP) into a shared 256-dimensional embedding space. The market and structure embeddings are summed to form a query ( $q = m + s$ ), which attends over the 24-month macro sequence via multi-head cross-attention ( $H = 4$ ). The resulting macro-conditioned representation  $c$  is passed through a classifier to produce a directional forecast and interpretable attention weights  $\alpha$  over macro lags.

### 3.1. Problem Setup

We consider a single asset observed over discrete trading days  $t = 1, \dots, T$ . For each time  $t$ , we observe three types of inputs:

- A sequence of market features over a lookback window of length  $L_{\text{mkt}}$  days, denoted

$$X_t^{\text{mkt}} \in \mathbb{R}^{L_{\text{mkt}} \times d_{\text{mkt}}},$$

where  $d_{\text{mkt}}$  is the number of market features (price, volume, technical indicators, and cross-asset signals).

- A sequence of macro features over a lookback window of length  $L_{\text{mac}}$  months, denoted

$$X_t^{\text{mac}} \in \mathbb{R}^{L_{\text{mac}} \times d_{\text{mac}}},$$

where  $d_{\text{mac}}$  is the number of macroeconomic variables (e.g., interest rates, inflation, employment, credit spreads). Daily macro observations are resampled to monthly frequency by taking the last value of each month.

- A vector of structure features capturing market microstructure characteristics, denoted

$$X_t^{\text{str}} \in \mathbb{R}^{d_{\text{str}}},$$

including volume-at-price (VAP) measures, anchored VWAP distances, gap statistics, and liquidity indicators.

Our goal is to predict the direction of asset returns over a forecast horizon of  $h$  days. We first define the  $h$ -day return as

$$r_{t,t+h} = \frac{P_{t+h} - P_t}{P_t},$$

where  $P_t$  denotes the closing price at time  $t$ . To focus on economically meaningful moves and reduce label noise, we exclude observations where the absolute return falls below a threshold  $\tau > 0$ :

Sample included if and only if  $|r_{t,t+h}| \geq \tau$ .

For the included samples, we define the binary label as

$$y_{t+h} = \begin{cases} 1 & \text{if } r_{t,t+h} > 0 \\ 0 & \text{if } r_{t,t+h} \leq 0 \end{cases}.$$

We aim to learn a mapping

$$f_{\theta} : (X_t^{\text{mkt}}, X_t^{\text{mac}}, X_t^{\text{str}}) \mapsto \hat{p}_{t+h} \in [0, 1],$$

parameterized by  $\theta$ , where  $\hat{p}_{t+h}$  represents the predicted probability that  $y_{t+h} = 1$ . The model is trained to minimize a composite loss combining binary cross-entropy with a differentiable AUC surrogate (Section 3.7).

### 3.2. Market Representation via Convolutional Network

The market branch processes the high-frequency technical indicator tensor  $X^{\text{mkt}} \in \mathbb{R}^{L_{\text{mkt}} \times d_{\text{mkt}}}$ , where  $L_{\text{mkt}} = 40$  trading days and  $d_{\text{mkt}} \approx 82$  denotes the number of selected features (varying slightly by fold due to per-fold feature selection). This tensor collects OHLCV-based technical indicators computed each day over the lookback window,



including momentum oscillators (RSI, Stochastic %K/%D), trend indicators (MACD, ADX), volatility measures (Bollinger Bands, ATR), and volume-based signals.

We use a one-dimensional convolutional neural network (1D-CNN) in which the temporal dimension is treated as the sequence length and the feature dimension is treated as the input-channel dimension. We first transpose the input to  $\mathbf{X}^{\text{mkt}} \in \mathbb{R}^{d_{\text{mkt}} \times L_{\text{mkt}}}$  so that convolution is applied along the time axis. The architecture consists of two convolutional blocks with channel progression  $[d_{\text{mkt}} \rightarrow 64 \rightarrow 128]$ :

$$\mathbf{H}^{(l)} = \text{Dropout}\left(\text{ReLU}\left(\text{BatchNorm}\left(\text{Conv1d}\left(\mathbf{H}^{(l-1)}\right)\right)\right)\right) \quad (1)$$

Each convolutional layer uses kernel size  $k = 3$  with padding chosen to preserve temporal resolution. Batch normalization is applied to improve training stability, followed by a ReLU activation. Dropout with probability  $p = 0.1$  is used for regularization.

To map the temporal dimension into a fixed-length representation, we use a dual global pooling scheme that concatenates global average pooling and global max pooling:

$$\mathbf{z}_{\text{pool}}^{\text{mkt}} = \left[ \text{AvgPool}(\mathbf{H}^{(L)}) \parallel \text{MaxPool}(\mathbf{H}^{(L)}) \right] \in \mathbb{R}^{256} \quad (2)$$

where  $\parallel$  denotes concatenation and  $\mathbf{H}^{(L)}$  has 128 channels after the final convolutional block. This pooling step preserves two different aspects of the temporal signal. Average pooling summarizes the overall level of activation across time, whereas max pooling retains the strongest responses, which often correspond to the most salient local patterns. The resulting 256-dimensional vector is then passed through a linear projection:

$$\mathbf{z}^{\text{mkt}} = \mathbf{W}_{\text{proj}} \mathbf{z}_{\text{pool}}^{\text{mkt}} + \mathbf{b}_{\text{proj}} \in \mathbb{R}^{256} \quad (3)$$

which yields the final market embedding  $\mathbf{z}^{\text{mkt}}$  with dimension  $d = 256$ .

### 3.3. Macro Representation via Temporal Convolutional Network

The macro branch processes a 24-month sliding window of macroeconomic indicators  $\mathbf{X}^{\text{mac}} \in \mathbb{R}^{24 \times d_{\text{mac}}}$ , where  $d_{\text{mac}} = 13$  features include the Federal Funds Rate, CPI, the unemployment rate, yield curve spreads, and other monthly economic indicators. We use a temporal convolutional network (TCN) to capture long-range dependence through dilated causal convolutions while preventing information leakage from future time steps [5].

We transpose the input to  $\mathbb{R}^{d_{\text{mac}} \times 24}$  so that 1D convolution is applied along the temporal axis. Our TCN contains  $L = 3$  stacked temporal blocks with a constant channel width  $C = 128$ . Each block  $b$  uses dilation factor  $d_b = 2^{b-1}$ , which gives dilations of 1, 2, and 4 across the three blocks.

Each temporal block contains two sequential dilated causal convolutions rather than a single convolution. For block  $b$  with input  $\mathbf{Z}^{(b-1)}$ , the forward pass is

$$\tilde{\mathbf{Z}}_1^{(b)} = \text{Dropout}\left(\text{ReLU}\left(\text{BatchNorm}\left(\text{Trim}\left(\text{Conv1D}_{d_b}\left(\mathbf{Z}^{(b-1)}\right)\right)\right)\right)\right) \quad (4)$$

$$\tilde{\mathbf{Z}}_2^{(b)} = \text{Dropout}\left(\text{ReLU}\left(\text{BatchNorm}\left(\text{Trim}\left(\text{Conv1D}_{d_b}\left(\tilde{\mathbf{Z}}_1^{(b)}\right)\right)\right)\right)\right) \quad (5)$$

where each convolution uses causal padding  $(k - 1) \cdot d_b$  followed by trimming back to the original sequence length, so that position  $t$  depends only on inputs at times  $\leq t$ . The block output includes a residual connection:

$$\mathbf{Z}^{(b)} = \text{ReLU}\left(\tilde{\mathbf{Z}}_2^{(b)} + \mathbf{W}_{\text{res}}\mathbf{Z}^{(b-1)}\right) \quad (6)$$

where  $\mathbf{W}_{\text{res}}$  is a  $1 \times 1$  convolution when channel dimensions differ; otherwise, it is an identity mapping.

After three temporal blocks, a final  $1 \times 1$  convolution projects to the fusion dimension:

$$\mathbf{H}^{\text{mac}} = \text{Conv1D}_{1 \times 1}(\mathbf{Z}^{(L)})^\top \in \mathbb{R}^{24 \times 256} \quad (7)$$

This yields a sequence of 24 monthly representations  $\mathbf{h}_j^{\text{mac}} \in \mathbb{R}^{256}$  for  $j = 1, \dots, 24$ , which serve as keys and values in the subsequent cross-attention mechanism. The exponentially growing receptive field enables capturing both short-term fluctuations and longer-term macroeconomic trends.

### 3.4. Structure Representation via MLP

The structure branch encodes the volume-at-price and microstructure feature vector  $X_t^{\text{str}} \in \mathbb{R}^{d_{\text{str}}}$  (where  $d_{\text{str}} = 24$ ) into the shared embedding space via a two-layer feedforward network:

$$\mathbf{z}_t^{\text{str}} = \mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 X_t^{\text{str}} + \mathbf{b}_1) + \mathbf{b}_2 \in \mathbb{R}^d, \quad (8)$$

where  $\mathbf{W}_1 \in \mathbb{R}^{64 \times d_{\text{str}}}$  and  $\mathbf{W}_2 \in \mathbb{R}^{d \times 64}$ . Unlike the market and macro branches, the structure input is a single vector per time step (not a temporal sequence), so no convolutional or recurrent architecture is needed.

### 3.5. Macro–Market Cross-Attention Fusion

To fuse market, structure, and macro information, we employ a cross-attention mechanism in which the combined market–structure representation acts as a query over the sequence of macro representations  $\mathbf{H}_t^{\text{mac}}$ . Intuitively, the model learns to ask: Which macro variables and historical lags are most relevant given the current market state and microstructure profile?

The query is formed by element-wise addition of the market and structure embeddings:

$$\mathbf{q}_t = \mathbf{m}_t + \mathbf{s}_t \in \mathbb{R}^d, \quad (9)$$

where  $\mathbf{m}_t \equiv \mathbf{z}_t^{\text{mkt}}$  is the market embedding from the CNN branch and  $\mathbf{s}_t \equiv \mathbf{z}_t^{\text{str}}$  is the structure embedding from the MLP branch (both in  $\mathbb{R}^d$ ,  $d = 256$ ).

We employ multi-head attention with  $H = 4$  heads. For each head  $h$ , we project the query, keys, and values into a lower-dimensional subspace of dimension  $d_h = d/H$ :

$$\begin{aligned} \mathbf{q}_t^{(h)} &= \mathbf{q}_t \mathbf{W}_Q^{(h)} \in \mathbb{R}^{d_h}, \\ \mathbf{K}_t^{(h)} &= \mathbf{H}_t^{\text{mac}} \mathbf{W}_K^{(h)} \in \mathbb{R}^{L_{\text{mac}} \times d_h}, \\ \mathbf{V}_t^{(h)} &= \mathbf{H}_t^{\text{mac}} \mathbf{W}_V^{(h)} \in \mathbb{R}^{L_{\text{mac}} \times d_h}, \end{aligned} \quad (10)$$

where  $\mathbf{W}_Q^{(h)}, \mathbf{W}_K^{(h)}, \mathbf{W}_V^{(h)} \in \mathbb{R}^{d \times d_h}$  are head-specific projection matrices.

The attention weights for head  $h$  are computed via scaled dot-product attention [7]:

$$\alpha_{t,\ell}^{(h)} = \frac{\exp((\mathbf{q}_t^{(h)})^\top \mathbf{k}_{t-\ell}^{(h)} / \sqrt{d_h})}{\sum_{j=0}^{L_{\text{mac}}-1} \exp((\mathbf{q}_t^{(h)})^\top \mathbf{k}_{t-j}^{(h)} / \sqrt{d_h})}, \quad (11)$$



where  $\mathbf{k}_{t-\ell}^{(h)}$  denotes the  $\ell$ -th row of  $\mathbf{K}_t^{(h)}$ . The per-head context vector is

$$\mathbf{c}_t^{(h)} = \sum_{\ell=0}^{L_{\text{mac}}-1} \alpha_{t,\ell}^{(h)} \mathbf{v}_{t-\ell}^{(h)} \in \mathbb{R}^{d_h}. \quad (12)$$

The head outputs are concatenated and passed through a linear projection:

$$\mathbf{c}_t = \text{Concat}(\mathbf{c}_t^{(1)}, \dots, \mathbf{c}_t^{(H)}) \mathbf{W}_O \in \mathbb{R}^d, \quad (13)$$

where  $\mathbf{W}_O \in \mathbb{R}^{d \times d}$  is the output projection matrix.

The attention weights  $\alpha_{t,\ell}^{(h)}$  reveal which macro lags the model deems relevant. We average across heads to obtain  $\bar{\alpha}_{t,\ell} = \frac{1}{H} \sum_{h=1}^H \alpha_{t,\ell}^{(h)}$ , which can be aggregated across time to study which historical macro periods most influence directional forecasts.

### 3.6. Prediction Head

The final prediction is computed from the macro-fused context vector  $\mathbf{c}_t$  via a feedforward classifier. Unlike architectures that concatenate multiple representations, our model relies solely on the cross-attention output, which already integrates market, structure, and macro information through the attention mechanism.

The classifier applies layer normalization followed by a two-layer MLP:

$$\begin{aligned} \tilde{\mathbf{c}}_t &= \text{LayerNorm}(\mathbf{c}_t), \\ \mathbf{h}_t &= \text{ReLU}(\mathbf{W}^{(1)} \tilde{\mathbf{c}}_t + \mathbf{b}^{(1)}), \\ \hat{y}_{t+h} &= \mathbf{w}^{(2)\top} \mathbf{h}_t + b^{(2)}, \end{aligned} \quad (14)$$

where  $\mathbf{W}^{(1)} \in \mathbb{R}^{64 \times d}$ ,  $\mathbf{w}^{(2)} \in \mathbb{R}^{64}$ , and dropout ( $p = 0.1$ ) is applied after the ReLU activation. The output  $\hat{y}_{t+h}$  represents the logit for the  $h$ -day-ahead directional forecast, with the probability of an upward move given by  $\sigma(\hat{y}_{t+h})$  where  $\sigma(\cdot)$  is the logistic sigmoid.

### 3.7. Training Objective

The model parameters  $\theta$  are learned by minimizing a composite loss that combines binary cross-entropy with a differentiable AUC surrogate. The cross-entropy component is

$$\mathcal{L}_{\text{BCE}}(\theta) = -\frac{1}{|\mathcal{D}_{\text{train}}|} \sum_{t \in \mathcal{D}_{\text{train}}} \left[ y_{t+h} \log \sigma(\hat{y}_{t+h}) + (1 - y_{t+h}) \log(1 - \sigma(\hat{y}_{t+h})) \right], \quad (15)$$

where  $y_{t+h} \in \{0, 1\}$  is the binary label,  $\hat{y}_{t+h}$  is the model's logit output, and  $\sigma(\cdot)$  is the logistic sigmoid.

To directly optimize ranking performance, we add a pairwise AUC surrogate loss [52]. Let  $\mathcal{P}$  and  $\mathcal{N}$  denote the sets of positive and negative samples in a mini-batch. The surrogate approximates the Wilcoxon–Mann–Whitney statistic via a smooth relaxation:

$$\mathcal{L}_{\text{AUC}}(\theta) = \frac{1}{|\mathcal{P}| |\mathcal{N}|} \sum_{p \in \mathcal{P}} \sum_{n \in \mathcal{N}} \log(1 + \exp(-(\hat{p}_p - \hat{p}_n))), \quad (16)$$

where  $\hat{p}_p = \sigma(\hat{y}_p)$  and  $\hat{p}_n = \sigma(\hat{y}_n)$  are the predicted probabilities for positive and negative samples, respectively. This softplus-based formulation provides smooth gradients when positive and negative predictions are close.

The total loss is a weighted combination:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{BCE}}(\theta) + \lambda \mathcal{L}_{\text{AUC}}(\theta), \quad (17)$$

with  $\lambda = 0.5$ . We use the AdamW optimizer with learning rate  $5 \times 10^{-4}$  and weight decay  $10^{-4}$ .

Training employs a 6-fold time-series cross-validation scheme with a 5-day embargo period between train and validation sets to prevent information leakage. Each fold uses an expanding window: the model trains on historical data and validates on a subsequent non-overlapping period. Early stopping with patience of 8 epochs monitors validation AUROC, and the learning rate is reduced by half when validation performance plateaus for 3 consecutive epochs.

### 3.8. Baseline Models and Ablation Variants

To assess the incremental value of each architectural component, we compare the proposed macro–market cross-attention model to a set of baseline and ablation variants. This section defines the main comparators at a high level; implementation details are provided in Section 3.10.

#### 3.8.1. Market-Only Model (Market-Only)

The first baseline uses only market features  $X_t^{\text{mkt}}$  and omits macro and structure inputs entirely. The architecture mirrors the market CNN branch described above, followed by a prediction head:

$$\hat{y}_{t+h}^{\text{Market-Only}} = f_{\text{cls}}(\mathbf{m}_t).$$

This model serves as a benchmark for assessing the incremental predictive content of macroeconomic and structural information.

#### 3.8.2. Macro-Only Model (Macro-Only)

To isolate the predictive power of macroeconomic information alone, we consider a macro-only model that applies mean pooling across the temporal dimension followed by a two-layer MLP:

$$\bar{\mathbf{x}}_t^{\text{mac}} = \frac{1}{L_{\text{mac}}} \sum_{\tau=1}^{L_{\text{mac}}} \mathbf{x}_{t,\tau}^{\text{mac}}, \quad \hat{y}_{t+h}^{\text{Macro-Only}} = f_{\text{cls}}(\text{MLP}(\bar{\mathbf{x}}_t^{\text{mac}})).$$

Although this model ignores price dynamics, it provides a useful reference for interpreting the standalone role of macro variables.

#### 3.8.3. Structure-Only Model (Structure-Only)

This baseline uses only VAP/volume profile features  $\mathbf{x}_t^{\text{str}}$  processed through a two-layer MLP:

$$\hat{y}_{t+h}^{\text{Structure-Only}} = f_{\text{cls}}(\text{MLP}(\mathbf{x}_t^{\text{str}})).$$

This isolates the predictive content of market microstructure features independent of price and macro dynamics.

#### 3.8.4. Concatenation Model (Concat)

This baseline incorporates all three input modalities via simple feature concatenation. Each branch uses the same encoder architecture as the main model, producing embeddings of dimension  $d = 256$ . The three embeddings are concatenated and fed into an MLP prediction head:

$$\mathbf{z}_t^{\text{concat}} = [\mathbf{m}_t; \bar{\mathbf{h}}_t^{\text{mac}}; \mathbf{s}_t], \quad \hat{y}_{t+h}^{\text{Concat}} = f_{\text{cls}}(\text{MLP}(\mathbf{z}_t^{\text{concat}})).$$

This baseline tests whether naive multi-modal fusion suffices without explicit cross-attention over macro lags.

### 3.8.5. Ablation: No Cross-Attention (NoCrossAttn)

In this variant, we retain all three branches—the market CNN, the macro TCN, and the structure MLP—but replace cross-attention with element-wise summation:

$$\mathbf{z}_t^{\text{sum}} = \mathbf{m}_t + \bar{\mathbf{h}}_t^{\text{mac}} + \mathbf{s}_t, \quad \hat{y}_{t+h}^{\text{NoCrossAttn}} = f_{\text{cls}}(\mathbf{z}_t^{\text{sum}}).$$

This ablation isolates the benefit of explicit macro–market cross-attention relative to simpler additive fusion mechanisms. By matching the encoder architectures exactly, performance differences can be attributed directly to the fusion strategy.

## 3.9. Time-Series Cross-Validation and Evaluation

Financial time series exhibit serial dependence and non-stationarity, requiring specialized cross-validation procedures to prevent look-ahead bias. We employ a 6-fold time-series cross-validation scheme with temporal embargo.

### 3.9.1. Fold Construction with Embargo

Let the full sample span  $[T_0, T_N]$ . For each fold  $k \in \{1, \dots, 6\}$ , we partition:

- Training set:  $\mathcal{D}_{\text{train}}^{(k)}$  containing samples from  $[T_{\text{start}}^{(k)}, T_{\text{end}}^{(k)}]$ .
- Embargo gap: 5 trading days to prevent information leakage from overlapping prediction horizons.
- Validation set:  $\mathcal{D}_{\text{val}}^{(k)}$  containing samples from  $[T_{\text{end}}^{(k)} + 5, T_{\text{val\_end}}^{(k)}]$ .

The embargo period ensures that no training sample's 5-day forward return window overlaps with any validation sample's feature window.

### 3.9.2. Evaluation Metrics

Given the binary classification formulation (predicting directional moves exceeding 1%), we evaluate models using AUROC (Area Under ROC Curve), which measures discrimination—the probability that a randomly chosen positive sample ranks higher than a randomly chosen negative. AUROC is threshold-free and serves as the primary metric for early stopping, model selection, and all reported results.

### 3.9.3. Training Protocol

Each fold trains for up to 60 epochs with early stopping (patience = 8 epochs) based on validation AUROC. We use ReduceLROnPlateau scheduling (factor 0.5, patience 3) and gradient clipping at 0.5. The best checkpoint by validation AUROC is retained for a final evaluation.

## 3.10. Implementation Details

All neural network components share a unified embedding dimension  $d = 256$ , ensuring compatibility across the fusion mechanism. The encoder architectures are detailed in Sections 3.2–3.4; here, we summarize key implementation choices.

The multi-head cross-attention module uses  $H = 4$  heads, which we found to be optimal in ablation studies (Section 7). Query, key, and value projections employ learned linear transformations without bias. Attention dropout of 0.1 is applied prior to the weighted sum of values to regularize the attention mechanism.

Models are trained with the AdamW optimizer using a learning rate of  $5 \times 10^{-4}$ . We employ a ReduceLROnPlateau scheduler with factor 0.5 and patience 3. Training runs

for up to 60 epochs with early stopping (patience 8) on validation AUROC, and the best checkpoint is retained for evaluation.

### Hyperparameter Optimization Protocol

Algorithm: Weight decay, dropout, and gradient clipping are tuned per model using exhaustive grid search rather than random or Bayesian search. Search space: The grid contains  $4 \times 4 \times 3 = 48$  configurations per model variant, with dropout  $\in \{0.05, 0.10, 0.15, 0.20\}$ , weight decay  $\in \{10^{-5}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$ , and gradient clipping  $\in \{0.5, 1.0, 2.0\}$ . Validation protocol: Each configuration is evaluated on the same 6-fold purged cross-validation splits used for reporting (Section 3.9); each fold trains an independent model on its training partition and scores out-of-fold AUROC, and the best configuration per model is selected by the mean AUROC across all six folds. Computational budget: With six model variants in the fair ablation (Section 7), this yields a total of  $6 \times 48 \times 6 = 1728$  training runs. Robustness of the selected configuration is assessed via the hyperparameter sensitivity analysis in Section 7, which shows that marginalizing over any single hyperparameter still leaves XAttnFusion above the best baseline. The best configuration for XAttnFusion is dropout = 0.10, weight decay =  $5 \times 10^{-4}$ , and gradient clipping = 2.0 (Table A3).

#### 3.11. Attention-Based Interpretability

A key feature of our cross-attention design is its built-in interpretability: the attention weights  $\alpha_{t,\ell}$  indicate which macro lags receive the most weight at prediction time  $t$ .

##### 3.11.1. Average Attention over Lags

For each sample  $t$ , the multi-head cross-attention mechanism produces attention distributions over  $L = 24$  monthly lags. We retain both the per-head weights  $\alpha_{t,\ell}^{(h)} \in \mathbb{R}^{H \times L}$  and the head-averaged counterpart:

$$\bar{\alpha}_{t,\ell} = \frac{1}{H} \sum_{h=1}^H \alpha_{t,\ell}^{(h)}, \quad \ell = 0, \dots, L-1, \quad (18)$$

where  $H = 4$  and  $\ell = 0$  correspond to the most recent macro observation. To summarize model behavior across the validation set, we compute the empirical mean attention:

$$\hat{\alpha}_\ell = \frac{1}{N} \sum_{t=1}^N \bar{\alpha}_{t,\ell}, \quad (19)$$

and plot it as a bar chart over lags (Section 6). Peaks indicate the historical macro horizons that receive the most consistent attention from the model.

##### 3.11.2. Attention Entropy

We measure how concentrated the attention weights are using entropy:

$$\mathcal{H}_t = - \sum_{\ell=0}^{L-1} \bar{\alpha}_{t,\ell} \log \bar{\alpha}_{t,\ell}. \quad (20)$$

Low entropy indicates that the model places most of its attention on a small set of lags, whereas high entropy indicates that attention is spread more broadly across the full macro history.

##### 3.11.3. Regime-Specific Attention Profiles

Each validation sample is assigned to one of three market regimes based on its proximity to high-volume nodes (HVN) or low-volume nodes (LVN) in the volume-at-price profile:

- Near-HVN: price within one standard deviation of the nearest HVN (support/resistance zone);
- LVN Corridor: price within one standard deviation of the nearest LVN (low-liquidity breakout zone);
- Neutral: neither condition applies.

For each regime  $r$ , we compute the average attention profile  $\hat{a}_\ell^{(r)}$  and compare regime-specific AUROC to assess whether attention patterns change systematically between relatively stable periods (near-HVN) and more volatile periods (near-LVN).

#### 3.11.4. Cross-Fold Stability Analysis

To assess whether the learned attention patterns are stable or specific to individual folds, we compare the average attention profiles  $\hat{a}_\ell^{(k)}$  across all  $K = 6$  cross-validation folds. High correlation across folds suggests stable macro-timing relationships, whereas divergent profiles may reflect changes in the underlying market environment.

These interpretability tools provide diagnostic evidence on the model's macro-timing behavior without relying on post hoc approximations such as SHAP or LIME, and they complement standard predictive performance metrics. We note that attention weights reflect learned statistical associations rather than causal relationships, and we interpret them as summaries of model behavior rather than evidence of economic mechanisms.

#### 3.12. Summary

To summarize, the proposed methodology combines:

1. A CNN-based market branch that extracts the local temporal structure from daily price and volume data.
2. An MLP-based structure branch that encodes volume-at-price (VAP) features capturing support/resistance levels.
3. A TCN-based macro branch that encodes long-range dependencies in macroeconomic time series into a sequence of macro representations.
4. A cross-attention fusion mechanism in which market and structure representations jointly query macro representations, yielding both a macro context vector for prediction and interpretable attention weights over macro lags.
5. A prediction head that maps the fused representation into directional forecasts, trained under an expanding-window cross-validation scheme with temporal embargo.
6. A set of baseline and ablation models that disentangle the contributions of macro inputs, cross-attention, and structure features.
7. Tools for analyzing attention weights to derive macro-timing insights and assess model stability across folds.

In the next section, we describe the empirical setting, including the target asset, forecast horizon, macro and market variables, and the data alignment procedures used in our experiments.

## 4. Data and Experimental Design

This section describes the empirical setting used to evaluate the proposed macro-market fusion architecture. We specify the target asset and forecast horizon, detail the construction and alignment of macroeconomic and market features, describe the volume-at-price structure features, and present the time-series cross-validation scheme.

#### 4.1. Target Asset and Forecast Horizon

Our primary analysis is applied to the SPDR S&P 500 ETF Trust (SPY), one of the most actively traded securities in global markets. We obtain market data for SPY and five related

instruments (QQQ (Nasdaq-100), IWM (Russell 2000), TLT (20+ Year Treasury), HYG (High Yield Corporate Bond), and VIXY (Short-Term VIX Futures)) from Yahoo Finance for the period January 2008 through December 2024. These auxiliary assets provide cross-asset signals for SPY prediction; in addition, we evaluate QQQ and IWM as separate prediction targets in Section 7.8.

The prediction target is the direction of the 5-day forward return, conditional on a minimum threshold. Formally, the simple return over horizon  $h = 5$  trading days is

$$r_{t,t+5} = \frac{P_{t+5} - P_t}{P_t},$$

where  $P_t$  denotes the closing price of SPY at time  $t$ . To reduce label noise from small and economically insignificant price moves, we exclude observations with  $|r_{t,t+5}| < 0.01$  and define the binary label as

$$y_{t+5} = \begin{cases} 1 & \text{if } r_{t,t+5} \geq 0.01, \\ 0 & \text{if } r_{t,t+5} \leq -0.01. \end{cases}$$

After filtering, the dataset contains 1802 unique prediction dates spanning March 2012 to December 2024. The out-of-fold validation period runs from February 2018 to December 2024 and contains 1865 sample-fold observations. Across validation folds, the base rate of positive labels is 62.63%, which reflects the long-run upward drift in U.S. equities over this period.

#### 4.2. Macroeconomic Data

We construct a panel of  $d_{\text{mac}} = 13$  macroeconomic variables from the Federal Reserve Economic Data (FRED) API, covering January 2004 to December 2024. The 4-year buffer before the market sample begins ensures that the 24-month macro lookback window is available at the earliest prediction date. The variables fall into five categories:

- Monetary policy and rates (2 variables): Federal Funds Rate (FEDFUNDS) and the yield curve slope (10-year minus 2-year Treasury spread, constructed from DGS10 and DGS2).
- Inflation (4 variables): CPI year-over-year change (CPIAUCSL), core CPI year-over-year (CPILFESL), PCE year-over-year (PCEPI), and core PCE year-over-year (PCEPILFE).
- Labor market (4 variables): nonfarm payrolls monthly change and year-over-year change (PAYEMS, two series), unemployment rate (UNRATE), and labor force participation rate (CIVPART).
- Credit conditions (2 variables): ICE BofA High Yield Option-Adjusted Spread (BAMLH0A0HYM2) and Chicago Fed National Financial Conditions Index (NFCI).
- Commodities (1 variable): WTI crude oil price (DCOILWTICO).

Macro series are published at varying frequencies (daily, weekly, monthly) and with publication lags. We align all series to daily frequency via forward-fill: each value persists until it is superseded by the next observation. To prevent look-ahead bias from publication delays (e.g., CPI and employment data are typically released two to six weeks after the observation period), we lag the macro window by two months: for a prediction date  $t$ , the most recent macro observation corresponds to month  $t - 2$ . The aligned and lagged panel is then resampled to monthly frequency by taking the last available value of each calendar month, yielding the macro input sequence  $X_t^{\text{mac}} \in \mathbb{R}^{24 \times 13}$ .

All macro variables are standardized using means and standard deviations computed on the training set of each fold. Outliers are winsorized at the 1st and 99th percentiles before standardization.



### Handling of Missing Values

Missing values in the macro panel are handled explicitly, and we never drop prediction dates on the basis of macro missingness:

- Gaps between releases. We forward-fill using last observation carried forward (LOCF), so the value of a slower-frequency series (e.g., monthly CPI) persists between releases. We do not interpolate linearly or with splines.
- Gaps before the first release of a series. Any residual missing values that remain after forward-filling (for example, early dates before a particular series has its first observation) are set to zero after per-fold standardization. Zero corresponds to the training-set mean of the standardized series and is therefore neutral with respect to the classifier.
- Sample dropping. We never drop prediction dates on the basis of macro missingness; all dates satisfying the market and structure availability requirements are retained.
- Availability mask. A binary availability mask is constructed alongside the macro input tensor  $X_t^{\text{mac}}$ , recording which monthly slots correspond to observations released before the cutoff date  $t - 2$ . The mask is passed to the cross-attention module and allows the model to distinguish observed entries from imputed ones.

### 4.3. Market Features

For each of the six assets and each trading day  $t$ , we compute a comprehensive set of market features from daily OHLCV data. These fall into four groups:

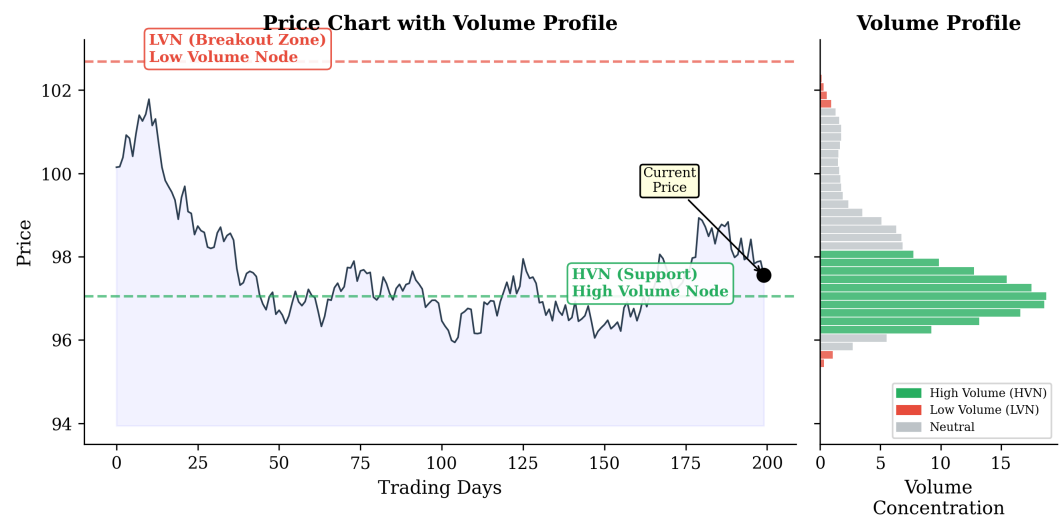
- Price and volume (per asset): open, high, low, close, volume, log returns, average true range (ATR, 14-period), volume-weighted average price (VWAP), overnight gaps, and Amihud illiquidity ratio.
- Technical indicators (per asset): momentum oscillators (RSI at 7 and 14 periods, Stochastic %K/%D, MACD line/signal/histogram, Rate of Change, Williams %R), trend indicators (ADX with directional indicators, EMA at 9/21/50 periods, SMA at 20/50 periods), volatility measures (Bollinger Band width and position, Keltner Channels, ATR percentage), and volume indicators (Money Flow Index, On-Balance Volume, Accumulation/Distribution, Chaikin Money Flow, Volume Price Trend).
- Cross-asset signals (6 features): SPY/QQQ price ratio, 5-day return differentials between SPY and QQQ, SPY and VIXY, and HYG and TLT (a proxy for credit conditions), 10-day return differential between SPY and TLT, and SPY/VIXY price ratio.
- Divergence signals: bullish and bearish divergence indicators derived from RSI, MACD, and OBV relative to price.

The raw feature set contains over 200 columns across six assets. To reduce dimensionality while retaining predictive information, we perform mutual-information-based feature selection separately within the training set of each fold. In each fold, features are ranked by their mutual information with the binary label, and the top 70 are retained. We then append 14 essential columns (SPY OHLCV, returns, ATR, VWAP, and the six cross-asset features), which are kept regardless of rank. This procedure yields roughly 82 market features per fold, with small variation across folds because of rank ties. The market input tensor is  $X_t^{\text{mkt}} \in \mathbb{R}^{40 \times d_{\text{mkt}}}$ , where 40 is the lookback window in trading days.

All market features are standardized within each fold using training-set statistics, and outliers are winsorized at the 1st and 99th percentiles.

#### 4.4. Structure Features: Volume-at-Price and Anchored VWAP

The structure branch takes a vector of  $d_{\text{str}} = 24$  features designed to capture aspects of market microstructure that are not contained in standard price and volume information (Figure 2). These features are computed from SPY data only and fall into four groups.



**Figure 2.** Volume-at-price (VAP) concept. (Left): A stylized price chart with horizontal dashed lines marking a high-volume node (HVN, green), where concentrated trading can create support, and a low-volume node (LVN, red), a thin region through which price often moves quickly. (Right): The corresponding volume profile histogram, where the bar color reflects the concentration of traded volume. The model's regime classification assigns each prediction date to near-HVN, LVN corridor, or neutral based on the current price's proximity to these nodes.

##### 4.4.1. Volume-at-Price (VAP) Profile Features (12 Features)

For each trading day  $t$ , we construct a volume-at-price profile over the preceding 6 months using kernel density estimation (KDE) with Gaussian bandwidth. From this profile, we identify high-volume nodes (HVN, local maxima) and low-volume nodes (LVN, local minima), which correspond to price levels of concentrated and thin trading activity, respectively. From the profile, we then compute the following 12 features: ATR-normalized distances from the current price to the nearest HVN and LVN (2), the price levels of the nearest HVN and LVN (2), a Herfindahl concentration index of the VAP distribution and the local concentration around the current price (2), the local slope and curvature of the VAP profile at the current price (2), the counts of HVNs and LVNs in the profile (2), and the total volume plus the price range of the profile (2). All VAP features are computed using only data available up to date  $t$ .

##### 4.4.2. Anchored VWAP Features (5 Features)

We identify rolling 20-day swing highs and lows in SPY price data and compute VWAPs anchored to the most recent swing high and swing low. Features include the percentage deviation in the current price from each anchored VWAP, the spread between the two anchored VWAPs (normalized by current price), and the number of trading days elapsed since the last swing high and swing low.

##### 4.4.3. Gap Features (5 Features)

We compute counts of up-gaps, down-gaps, filled gaps, and unfilled large gaps ("air pockets") over the most recent 5 trading days.

#### 4.4.4. Liquidity Features (2 Features)

The Amihud illiquidity ratio, defined as absolute return divided by dollar volume, and its 20-day rolling average capture changes in market liquidity over time.

The structure input is a single vector  $X_t^{\text{str}} \in \mathbb{R}^{24}$ , standardized within each fold using training-set statistics.

#### 4.5. Sample Period and Preprocessing

The market data cover January 2008 to December 2024, but the effective sample begins in March 2012 because the model requires a 40-day market lookback window, a 24-month macro lookback window, and a 6-month VAP lookback window. After applying the minimum-move filter ( $|r_{t,t+5}| \geq 1\%$ ), the sample contains 1802 unique prediction dates.

For each date, the model receives three aligned inputs: the market tensor  $X_t^{\text{mkt}} \in \mathbb{R}^{40 \times 82}$ , the macro tensor  $X_t^{\text{mac}} \in \mathbb{R}^{24 \times 13}$  together with a binary mask indicating which monthly observations are available, and the structure vector  $X_t^{\text{str}} \in \mathbb{R}^{24}$ .

Preprocessing follows a strict temporal protocol. All feature transformations, including returns, technical indicators, and VAP profiles, use only information available up to the prediction date. All normalization statistics, including means, standard deviations, and winsorization quantiles, are computed from the training partition of each fold. Missing values remaining after forward-fill are set to zero (neutral after standardization).

#### 4.6. Cross-Validation Design

We evaluate all models using a 6-fold purged cross-validation scheme with a 5-day embargo, as described in Section 3.9. Table 1 reports the fold structure.

**Table 1.** Cross-validation fold structure.

| Fold | Train Period                | Val Period                  | Train $n$ | Val $n$ | Val UP% |
|------|-----------------------------|-----------------------------|-----------|---------|---------|
| 1    | January 2017–January 2023   | January 2023–December 2024  | 876       | 284     | 67.96   |
| 2    | January 2016–January 2022   | January 2022–January 2024   | 798       | 330     | 55.76   |
| 3    | February 2015–January 2021  | January 2021–January 2023   | 804       | 321     | 56.07   |
| 4    | February 2014–January 2020  | February 2020–January 2022  | 741       | 319     | 65.52   |
| 5    | February 2013–February 2019 | February 2019–February 2021 | 740       | 323     | 67.80   |
| 6    | February 2012–February 2018 | February 2018–February 2020 | 727       | 288     | 63.54   |

Notes: Each fold uses an expanding training window. A 5-day embargo separates the last training date from the first validation date. Val UP% is the fraction of positive labels in the validation set. The total number of out-of-fold validation observations is 1865.

Each fold trains on an expanding historical window and validates on a subsequent non-overlapping period. The 5-day embargo ensures that no training sample's 5-day forward return window overlaps with any validation sample's feature window. Because the expanding window produces overlapping validation periods across folds, a given date may appear in multiple folds' validation sets; however, each fold trains an independent model on distinct historical data, so per-fold AUROC estimates are computed from separately trained models. Feature selection (mutual information) and normalization are performed independently within each fold's training partition, so that no information from the validation period leaks into model estimation.

Within each fold, we train for up to 60 epochs with early stopping (patience 8) based on validation AUROC. The best checkpoint is retained for evaluation. Hyperparameter tuning is conducted via a grid search over dropout, weight decay, and gradient clipping, with each model variant tuned independently to ensure a fair comparison (Section 7).

## 5. Empirical Evaluation

This section reports the out-of-sample predictive performance of XAttnFusion relative to baseline models. We use discrimination, measured by AUROC, as the primary evaluation metric because it is threshold-free and well suited to imbalanced binary classification with a 62.63% base rate.

### 5.1. Main Results

Table 2 reports the performance of XAttnFusion under the best configuration identified by the 48-configuration hyperparameter search (dropout = 0.1, weight decay =  $5 \times 10^{-4}$ , gradient clipping = 2.0). All macro inputs are lagged by two months to prevent look-ahead bias. Across the six cross-validation folds, the model achieves a mean out-of-fold AUROC of  $0.6333 \pm 0.0473$ , with fold-specific values ranging from 0.568 to 0.705.

**Table 2.** XAttnFusion out-of-fold performance (best configuration from grid search).

| Metric | Fold 1 | Fold 2 | Fold 3 | Fold 4 | Fold 5 | Fold 6 | Mean $\pm$ Std    |
|--------|--------|--------|--------|--------|--------|--------|-------------------|
| AUROC  | 0.639  | 0.680  | 0.597  | 0.705  | 0.568  | 0.611  | $0.633 \pm 0.047$ |

Notes: AUROC is computed on the out-of-fold validation set for each purged cross-validation fold with a 5-day embargo.

AUROC stays above 0.56 in every fold, indicating that the model preserves a ranking signal across a range of market environments from 2018 to 2024, including the 2020 COVID-19 crash and the 2022 rate-hiking cycle. The weakest performance occurs in fold 5 (validation 2019–2021, AUROC 0.568), whose window contains the emergency rate cuts and pandemic disruption, episodes in which rapid macro regime shifts reduce the value of lagged macro information.

### 5.2. Baseline Comparison

To place XAttnFusion's performance in context, we compare it against four baseline models trained on the same multi-modal inputs (market, macro, and structure) and evaluated on the same cross-validation splits. We also include a buy-and-hold benchmark that always predicts the positive class and thus matches the base rate by construction.

Table 3 shows that XAttnFusion delivers the highest AUROC among all models by a substantial margin. The best baselines, TFT-Lite-Fusion and Logistic Regression (both 0.559), fall 0.074 AUROC points below XAttnFusion (0.633). LightGBM performs at chance (0.496), illustrating that standard tabular learners cannot extract a predictive signal from lagged macroeconomic sequences without temporal modeling.

**Table 3.** Baseline comparison (all models receive the same multi-modal inputs).

| Model                    | Mean AUROC | Std   |
|--------------------------|------------|-------|
| LightGBM                 | 0.496      | 0.027 |
| Buy-and-Hold (always UP) | 0.500      | 0.000 |
| XGBoost                  | 0.548      | 0.071 |
| LSTM-Fusion              | 0.554      | 0.062 |
| Logistic Regression      | 0.559      | 0.034 |
| TFT-Lite-Fusion          | 0.559      | 0.049 |
| XAttnFusion              | 0.633      | 0.047 |

Notes: All models receive the same market, macro, and structure inputs (with 2-month macro lag) and are evaluated on the same 6-fold purged CV splits. Buy-and-Hold always predicts the majority class (AUROC = 0.50). XGBoost, LightGBM, and Logistic Regression use flattened summary statistics of the input tensors; LSTM-Fusion and TFT-Lite-Fusion are neural architectures with comparable fusion designs.

The model is not intended to serve as a trading strategy, and economic evaluation through portfolio backtesting is left for future work. The task is directional classification with a 1% minimum-move filter, not direct return forecasting. An AUROC of 0.633 means that given a random positive and negative sample, the model ranks them correctly 63.3% of the time. This represents a modest but consistent signal that could inform signal weighting or risk overlay in a broader investment process, though it is insufficient as a standalone trading signal. Our focus is the scientific question of whether cross-attention fusion improves macro–market discrimination, and whether that improvement varies with the market state, a question we examine further in Section 6.

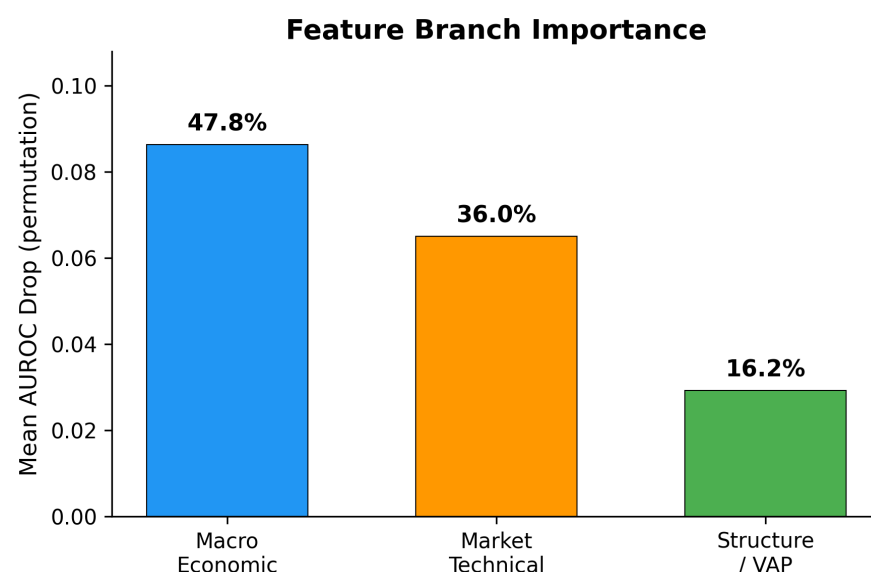
## 6. Interpretability and Regime Analysis

This section presents the paper’s main empirical result: the model’s predictive power depends on market structure regime and changes systematically across states. We first quantify the relative contribution of each input branch using permutation-based feature importance and then examine how discrimination changes across volume-at-price regimes.

### 6.1. Feature Branch Importance

To assess the relative contribution of each input modality, we compute permutation-based feature importance at the branch level. For each branch (macro, market, structure), we permute the branch’s inputs across samples and measure the resulting drop in out-of-fold AUROC. Table 4 reports the results.

Figure 3 visualizes these results. Macro variables account for 48% of the model’s predictive importance, followed closely by market technical features at 36%, while structure features contribute 16%. Because macro inputs are restricted to information available two months before the prediction date, they retain substantial but not dominant predictive weight; market technical features, which reflect short-term price dynamics, contribute on a comparable scale. This balance suggests that XAttnFusion’s advantage over concatenation does not stem from amplifying a single dominant input, but from its ability to fuse two comparably weighted sources: the market state selectively queries the macro sequence rather than averaging macro and market representations blindly. Structure features, though a modest share of overall importance, play a critical role in determining when the model’s predictions are reliable, as we show below.



**Figure 3.** Feature branch importance measured by AUROC drop when each branch’s inputs are permuted. Macro variables contribute 48%, market technical features 36%, and structure features 16%.

**Table 4.** Feature branch importance (permutation-based AUROC drop).

| Feature Branch   | AUROC Drop | Importance % |
|------------------|------------|--------------|
| Macro Economic   | 0.0863     | 47.8         |
| Market Technical | 0.0650     | 36.0         |
| Structure/VAP    | 0.0293     | 16.2         |

Notes: Importance is computed by permuting the inputs to each branch and measuring the mean drop in AUROC across validation folds. Percentages sum to 100 and indicate relative contributions.

### 6.2. State-Dependent Predictive Power

A central finding of this paper is that the model's discriminative ability varies systematically with market conditions, depending on the price's proximity to a volume-at-price structure. Each validation sample is assigned to one of three regimes based on the z-score-normalized distance to the nearest high-volume node (HVN) and low-volume node (LVN):

- **Near-HVN:** The price lies within one standard deviation of the nearest high-volume node, which corresponds to a region of historically concentrated trading that often functions as support or resistance.
- **LVN Corridor:** The price lies within one standard deviation of the nearest low-volume node, a thin region with limited historical trading that is often linked to breakouts or rapid transitions.
- **Neutral:** Neither condition holds, so the price lies in an intermediate region away from the main volume features.

Table 5 reports regime-specific performance, computed per fold and averaged across all six folds.

**Table 5.** Regime-specific predictive performance (per-fold averaged).

| Regime       | N   | % of Data | Mean AUROC | Std   | Base Rate |
|--------------|-----|-----------|------------|-------|-----------|
| Near-HVN     | 671 | 36.0      | 0.732      | 0.111 | 62.0%     |
| LVN Corridor | 810 | 43.4      | 0.550      | 0.056 | 67.2%     |
| Neutral      | 384 | 20.6      | 0.710      | 0.117 | 61.2%     |

Notes: Regimes are classified by z-score-normalized distance to the nearest HVN and LVN in the volume-at-price profile (threshold =  $-0.5$  z-score). AUROC is computed within each fold separately and then averaged, since each fold trains an independent model. Standard deviations reflect cross-fold variability. Per-regime sample sizes of 64–135 per fold contribute to high variance in the near-HVN and neutral estimates.

The results reveal state-dependent predictive power. The most robust finding is that the model performs worst in LVN corridors (mean AUROC  $0.550 \pm 0.056$ ), the regime with the largest sample size and lowest variance. Performance is higher near high-volume nodes ( $0.732 \pm 0.111$ ) and in neutral zones ( $0.710 \pm 0.117$ ), though these estimates carry greater uncertainty due to smaller per-fold sample sizes ( $\sim 112$  and  $\sim 64$  samples per fold, respectively). The spread between the best and worst regime averages 0.18 in AUROC, confirming that the value of macro–market fusion varies substantially with market structure.

### 6.3. Economic Interpretation

The regime-dependent pattern admits a natural economic interpretation.

#### 6.3.1. LVN Corridor (Breakout/Transition Zones)

Low-volume nodes correspond to price levels that the market has passed through quickly, leaving relatively little accumulated positioning behind. In LVN corridors, price movements are driven more by momentum, stop-loss cascades, and short-horizon order-flow imbalances than by macroeconomic fundamentals. As a result, the cross-attention mechanism, which is designed to extract macro-conditioned signals, has less predictive



traction in this regime. The consistently lower AUROC in LVN corridors ( $0.550 \pm 0.056$ ) is the most stable regime-level finding and supports the view that macro-market fusion adds less value when price dynamics are momentum-driven.

### 6.3.2. Near-HVN (Support/Resistance Zones)

High-volume nodes mark price levels where two-way trading has historically been concentrated. When prices approach these levels, the next move may depend more on fundamental conditions, allowing the cross-attention mechanism to extract useful macro signals. The higher mean AUROC near HVNs (0.732) is consistent with this interpretation, though the wide cross-fold variation ( $\text{std} = 0.111$ ) warrants caution.

### 6.3.3. Neutral Zones

Performance in neutral zones ( $0.710 \pm 0.117$ ) is comparable to near-HVN but highly variable across folds (per-fold AUROC ranges from 0.579 to 0.881). A bootstrap test (10,000 resamples of the 6 fold-level AUROCs) confirms that the neutral mean is significantly above chance ( $p < 0.001$ ; 95% CI [0.616, 0.808]). However, with only  $\sim 64$  samples per fold in this regime, the wide confidence interval warrants caution in interpreting the point estimate.

### 6.3.4. Robustness to Alternative Regime Definition

To test whether the state-dependent finding is specific to the VAP-based regime classification, we repeat the analysis using an alternative regime definition based on 20-day realized volatility. Validation dates are split into terciles (low, mid, and high volatility). The qualitative pattern is consistent: the model discriminates best in low-volatility (mean AUROC 0.641) and high-volatility (0.647) environments and worst in mid-volatility periods (0.587), mirroring the Near-HVN/LVN contrast. This suggests that the regime dependence of predictive performance is not an artifact of the particular VAP-based classification but reflects a more general property of market conditions.

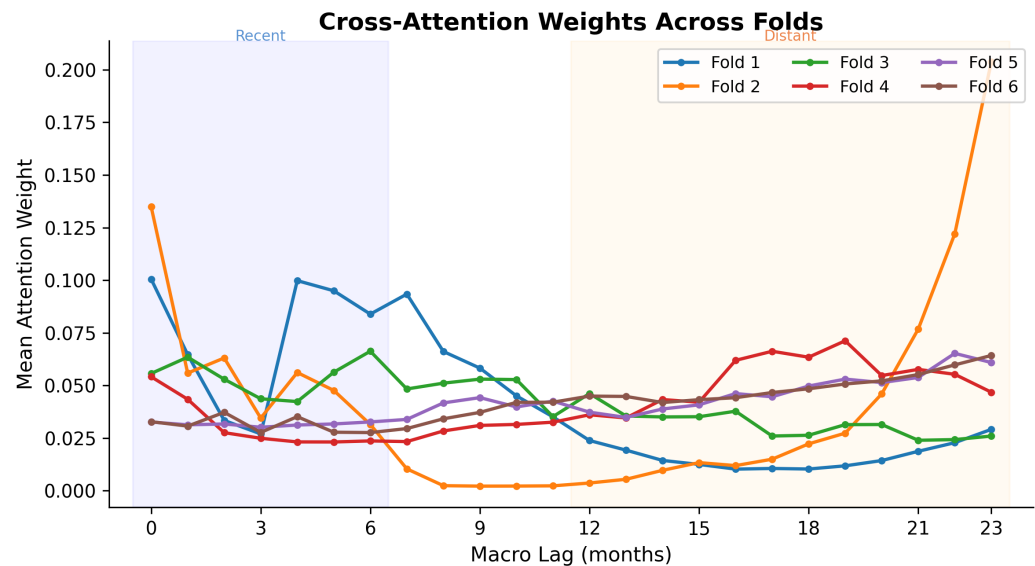
## 6.4. Attention Patterns Across Regimes

To further characterize regime dependence, we examine how the cross-attention mechanism distributes weight across the 24 monthly macro lags within each regime. For each validation sample, we record the head-averaged attention vector  $\bar{\alpha}_{t,\ell}$  and compute several regime-level summary statistics: the center of mass (the average attended lag), entropy (the concentration of attention), and the share of attention assigned to recent lags (months 0–6) relative to distant lags (months 12–24).

Figure 4 plots the mean attention weights across the 24 macro lags, conditioned on regime and averaged over folds.

Attention profiles vary meaningfully across folds in a way that tracks the macro environment of each validation window. In fold 1, which validates the relatively stable 2023–2024 period after the Federal Reserve’s tightening cycle, the center of mass sits near lag 8 and the model relies predominantly on recent macro history (the most recent six months). In the remaining five folds, whose validation windows span macroeconomically disruptive episodes—the 2018–2019 monetary policy inflection and the onset of the 2020 COVID-19 crash (fold 6), the COVID-19 crash and recovery (folds 4 and 5), and the 2021–2022 inflation surge and early tightening phase (folds 2 and 3)—the center of mass shifts to lag 13–14 and the majority of attention falls on distant lags ( $>12$  months). This pattern is consistent with the intuition that the model leans on older macro context when recent conditions are less representative of the longer-run training distribution. Across all six folds, the cross-fold average center of mass is  $12.2 \pm 2.7$  for near-HVN samples,  $11.8 \pm 2.4$  for LVN corridor, and  $9.9 \pm 1.6$  for neutral, with mean entropy  $\approx 2.99$  across regimes.





**Figure 4.** Cross-attention weights over 24 monthly macro lags for each of the 6 cross-validation folds. Blue shading marks recent lags (0–6 months); yellow marks distant lags (12–24 months). Attention profiles vary meaningfully across folds: in fold 1 (validating the relatively stable 2023–2024 period) the center of mass sits near lag 8 and the model relies on recent macro history, while in the other folds—whose validation windows span COVID-19, the recovery, and the tightening cycle—the center of mass shifts to lag 13–14 and attention concentrates on distant lags.

Within folds, regime-specific differences in attention are modest. The centers of mass and entropy are similar between near-HVN and LVN samples, indicating that the regime-dependent performance gap is driven primarily by which samples the model encounters, not by large shifts in the attention mechanism itself. The cross-attention mechanism provides a stable macro summary; it is the interaction of that summary with market structure that determines predictive success.

## 7. Ablation Analysis

To isolate the contribution of each architectural component, we conduct a fair ablation study in which every model variant is independently hyperparameter-tuned over the same search space. This ensures that differences in performance reflect the architecture itself rather than an advantage in hyperparameter tuning.

### 7.1. Fair Ablation Design

We compare six model variants, each tuned over a grid of 48 configurations (dropout  $\in \{0.05, 0.10, 0.15, 0.20\}$ , weight decay  $\in \{10^{-5}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$ , gradient clipping  $\in \{0.5, 1.0, 2.0\}$ ). Table 6 reports the AUROC achieved by the best configuration for each variant, along with a paired *t*-test and Wilcoxon *p*-values against XAttnFusion.

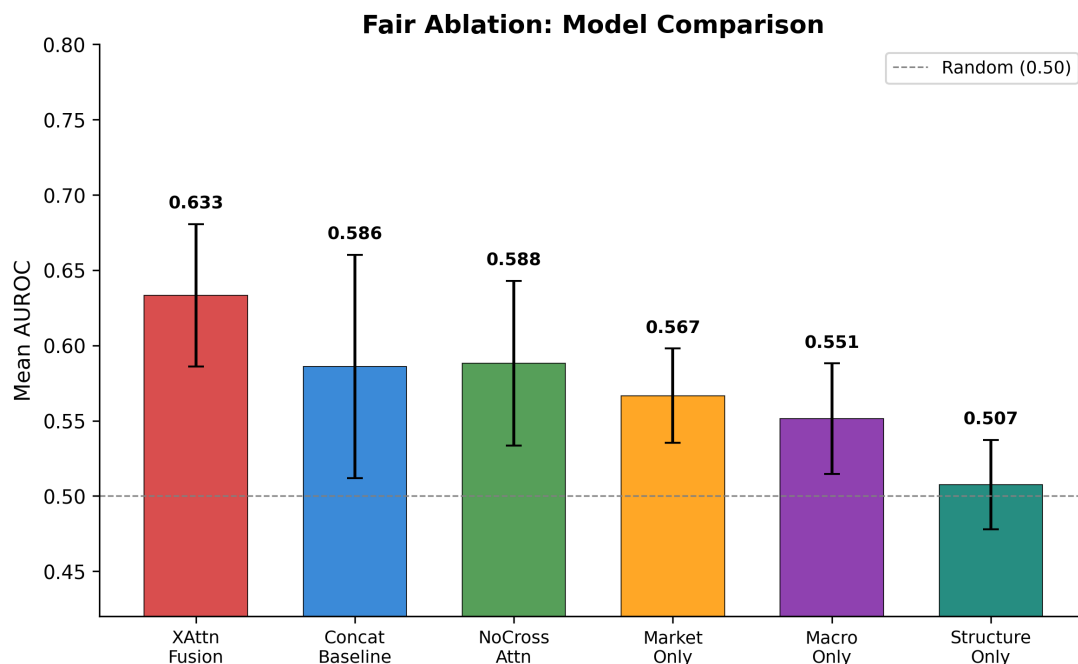
**Table 6.** Fair ablation results (each model independently tuned).

| Model          | Mean AUROC | Std    | $\Delta$ vs. XAttn | <i>p</i> ( <i>t</i> -Test) | <i>p</i> (Wilcoxon) |
|----------------|------------|--------|--------------------|----------------------------|---------------------|
| XAttnFusion    | 0.6333     | 0.0473 | —                  | —                          | —                   |
| ConcatBaseline | 0.5861     | 0.0742 | −0.0472            | 0.111                      | 0.031               |
| NoCrossAttn    | 0.5881     | 0.0547 | −0.0452            | 0.185                      | 0.156               |
| MacroOnly      | 0.5513     | 0.0368 | −0.0820            | 0.036                      | 0.031               |
| MarketOnly     | 0.5666     | 0.0314 | −0.0667            | 0.004                      | 0.031               |
| StructureOnly  | 0.5074     | 0.0297 | −0.1259            | 0.005                      | 0.031               |

Notes: Each model is tuned over 48 hyperparameter configurations and evaluated on the same 6-fold purged CV splits with 2-month macro lag.  $\Delta$  is the AUROC difference relative to XAttnFusion. We report both parametric (paired *t*-test) and non-parametric (Wilcoxon signed-rank) *p*-values given the small number of folds ( $n = 6$ ).

### 7.2. Cross-Attention vs. Alternative Fusion

Figure 5 visualizes the results. The most important comparison is between XAttnFusion and models that use the same inputs but different fusion strategies.



**Figure 5.** Fair ablation comparison. Each model is tuned independently over 48 hyperparameter configurations. Error bars report the standard deviation across 6 folds. XAttnFusion (red) attains the highest AUROC, and the dashed line marks random performance.

The ConcatBaseline, which concatenates market, macro (mean-pooled), and structure embeddings, achieves 0.5861, an AUROC gap of  $-0.0472$  relative to XAttnFusion. The Wilcoxon signed-rank test yields  $p = 0.031$ , significant at the 5% level, while the paired  $t$ -test gives  $p = 0.111$ , reflecting lower power with only six folds. The NoCrossAttn variant, which sums the three embeddings element-wise, achieves 0.5881 ( $\Delta = -0.0452$ ). Both comparisons indicate that the cross-attention mechanism, which allows the market state to selectively query the macro sequence, captures information that simpler fusion strategies miss.

### 7.3. Component Contributions

A comparison of the single-branch models shows a clear ranking. MarketOnly (0.5666) outperforms MacroOnly (0.5513), indicating that market technical features carry more standalone predictive power than macro variables alone when macro inputs are lagged by two months. StructureOnly performs near chance (0.5074), suggesting that structure features alone do not carry a directional signal. Their role is instead to provide conditioning information that shapes the effectiveness of macro–market fusion, as discussed in Section 6.2.

### 7.4. Structure Feature Ablation

To measure the incremental contribution of structure features within the full model, we compare XAttnFusion with and without the structure branch by zeroing the structure inputs at inference time using the tuned checkpoints. Structure features improve AUROC in 5 of the 6 folds, with a mean gain of  $+0.014$ . The improvement is modest, but it is consistent, especially given that the structure branch accounts for only 16% of total feature importance. Its main contribution is not direct predictive power. Instead, it provides regime context

that helps the model concentrate its discrimination in market states where the signal is more informative.

### 7.5. Hyperparameter Sensitivity

Table 7 reports how sensitive XAttnFusion is to the main hyperparameters explored in the grid search.

**Table 7.** Hyperparameter sensitivity (XAttnFusion).

| Configuration                                  | Mean AUROC | Std    |
|------------------------------------------------|------------|--------|
| Dropout = 0.05                                 | 0.5970     | 0.0074 |
| Dropout = 0.10 <sup>†</sup>                    | 0.6077     | 0.0121 |
| Dropout = 0.15                                 | 0.5992     | 0.0085 |
| Dropout = 0.20                                 | 0.5961     | 0.0103 |
| Weight decay = $10^{-5}$                       | 0.5980     | 0.0090 |
| Weight decay = $10^{-4}$                       | 0.6032     | 0.0094 |
| Weight decay = $5 \times 10^{-4}$ <sup>†</sup> | 0.5976     | 0.0135 |
| Weight decay = $10^{-3}$                       | 0.6012     | 0.0099 |
| Grad clip = 0.5                                | 0.6006     | 0.0053 |
| Grad clip = 1.0                                | 0.5986     | 0.0092 |
| Grad clip = 2.0 <sup>†</sup>                   | 0.6008     | 0.0152 |

Notes: Each row reports the mean AUROC marginalized over all configurations sharing that hyperparameter value (12 configs per dropout/weight-decay row, 16 per grad-clip row). Rows marked with <sup>†</sup> indicate values that belong to the single best joint configuration selected by grid search (dropout = 0.10, weight decay =  $5 \times 10^{-4}$ , grad clip = 2.0, mean AUROC = 0.6333). Because marginal and joint optima can differ, a marked row may not have the highest marginal mean: for example, weight decay =  $5 \times 10^{-4}$  has a lower marginal mean than  $10^{-4}$ , yet it produces the best performance in combination with dropout = 0.10 and grad clip = 2.0. All marginalized averages exceed 0.59, above the best baseline (ConcatBaseline, 0.586), indicating that the model's advantage is not driven by narrow tuning.

AUROC stays within a relatively narrow range across the hyperparameter grid. The model does not appear fragile: the relative ranking of model components remains the same across settings.

### 7.6. Threshold Sensitivity

To assess sensitivity to the minimum-move filter, we re-run the full fair ablation at thresholds of 0.5% and 2% in addition to the default 1%. Table 8 reports XAttnFusion and ConcatBaseline AUROC at each threshold.

**Table 8.** Threshold sensitivity: XAttnFusion vs. ConcatBaseline.

| Threshold      | XAttnFusion | ConcatBaseline | $\Delta$ |
|----------------|-------------|----------------|----------|
| 0.5%           | 0.6027      | 0.5490         | +0.054   |
| 1.0% (default) | 0.6333      | 0.5861         | +0.047   |
| 2.0%           | 0.6109      | 0.6374         | −0.026   |

Notes: Each row re-runs the full 48-configuration fair ablation on feature stores generated with the indicated threshold. All experiments use 2-month macro lag.

Cross-attention fusion outperforms concatenation at the 0.5% and 1% thresholds but not at 2%. At the 2% threshold, the prediction task is restricted to large moves that are more likely driven by momentum and order-flow dynamics than by macroeconomic fundamentals. In this regime, the cross-attention mechanism adds less value because the macro-conditioned signal it extracts is less relevant. This pattern is consistent with the state-dependent finding in Section 6.2: the benefit of macro–market fusion depends on the nature of the prediction task.

### 7.7. Feature Count Sensitivity

To verify that results are not sensitive to the choice of retaining 70 features via mutual information, we re-run the fair ablation for  $k \in \{50, 70, 90\}$ . Table 9 reports XAttnFusion AUROC at each setting.

**Table 9.** Feature count sensitivity (top- $k$  MI features).

| Top- $k$     | XAttnFusion | ConcatBaseline | $\Delta$ |
|--------------|-------------|----------------|----------|
| 50           | 0.6289      | 0.6057         | +0.023   |
| 70 (default) | 0.6333      | 0.5861         | +0.047   |
| 90           | 0.6167      | 0.6066         | +0.010   |

Notes: Each row re-runs the full 48-configuration fair ablation with the indicated number of MI-selected features. The 14 essential columns are always included.

XAttnFusion outperforms ConcatBaseline at all three feature counts. Performance is highest at  $k = 70$  and degrades slightly at  $k = 90$ , suggesting that additional low-MI features introduce noise. The cross-attention advantage is robust to this design choice.

### 7.8. Multi-Asset Evaluation

To test whether the findings generalize beyond SPY, we apply the same pipeline to QQQ (Nasdaq-100 ETF) and IWM (Russell 2000 ETF) as prediction targets. Structure features (VAP, anchored VWAP) are recomputed from each target asset's price and volume data; macro and cross-asset features remain the same. Table 10 reports the results.

**Table 10.** Multi-asset evaluation: cross-attention benefit across target assets.

| Target | XAttnFusion | ConcatBaseline | NoCrossAttn | $\Delta$ vs. Concat |
|--------|-------------|----------------|-------------|---------------------|
| SPY    | 0.6333      | 0.5861         | 0.5881      | +0.047              |
| QQQ    | 0.6095      | 0.5867         | 0.5828      | +0.023              |
| IWM    | 0.6260      | 0.5812         | 0.5612      | +0.045              |

Notes: Each row uses the same 48-configuration fair ablation with 2-month macro lag. Feature stores are generated independently for each target asset.

XAttnFusion outperforms both ConcatBaseline and NoCrossAttn on all three assets. The cross-attention benefit is largest for SPY (+0.047) and IWM (+0.045) and more modest for QQQ (+0.023). These results suggest that the advantage of macro–market cross-attention fusion is not specific to a single asset but reflects a more general property of how macroeconomic information interacts with market dynamics.

## 8. Conclusions

This paper introduces XAttnFusion, a macro–market fusion architecture that combines a CNN market branch, a TCN macro branch, and a structure-aware feedforward branch through multi-head cross-attention. We evaluate the model on a 5-day directional prediction task for the S&P 500 ETF (SPY) over the 2012–2024 period using purged cross-validation with a 5-day embargo.

The main findings are:

1. With a 2-month macro lag to prevent look-ahead bias from publication delays, XAttnFusion achieves a mean out-of-fold AUROC of  $0.6333 \pm 0.0473$ , exceeding all baselines by at least 0.074 AUROC points.
2. In a fair ablation in which each model is tuned independently, cross-attention fusion improves AUROC by +0.047 relative to concatenation ( $p = 0.031$ , Wilcoxon signed-rank test). The model generalizes beyond SPY: cross-attention consistently

outperforms alternative fusion strategies on QQQ (Nasdaq-100) and IWM (Russell 2000) as well.

3. The cross-attention advantage is robust to the minimum-move threshold (0.5–1%) and the number of MI-selected features ( $k = 50\text{--}90$ ), but diminishes at the 2% threshold where large moves are driven more by momentum than macroeconomic fundamentals.
4. Permutation-based feature importance shows that macro (48%) and market (36%) branches contribute on a comparable scale. XAttnFusion's gain over concatenation therefore comes from learning how to fuse two comparably weighted sources rather than amplifying a single dominant input.
5. The model's predictive power is state-dependent, indicating that the value of macro–market fusion is itself conditioned on market structure.

### 8.1. Limitations

This study has several limitations. Although we extend the evaluation to QQQ and IWM, the analysis remains limited to U.S. equity ETFs; generalization to other asset classes or geographic settings is untested. The model predicts direction rather than magnitude and is neither designed nor evaluated as a trading system; economic evaluation through portfolio backtesting is left for future work. Although a volatility-based robustness check confirms the qualitative regime pattern, the regime classification relies on simple z-score thresholds, and more sophisticated regime definitions (e.g., hidden Markov models or learned latent states) may yield different empirical patterns.

### 8.2. Future Directions

Several extensions remain for future research:

- Alternative regime definitions based on volatility regimes, macro regimes, or learned latent states.
- Temporal attention analysis to examine which macro lags receive attention in different regimes.

More broadly, our results suggest that, in financial deep learning, it is not enough to ask whether a model predicts well. It is equally important to understand when and where its predictions contain useful signals, especially in relation to market structure and macroeconomic conditions.

**Author Contributions:** Conceptualization, J.R., S.J. and Z.Z.; Methodology, J.R., S.J. and Z.Z.; Software, J.R. and S.J.; Validation, J.R. and S.J.; Formal analysis, J.R., S.J. and Z.Z.; Investigation, J.R., S.J. and Z.Z.; Data curation, J.R. and S.J.; Writing—original draft, J.R., S.J. and Z.Z.; Writing—review & editing, J.R., S.J. and Z.Z.; Visualization, J.R. and S.J.; Supervision, Z.Z. All authors have read and agreed to the published version of the manuscript.

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**Data Availability Statement:** The data presented in this study are openly available in Yahoo Finance at <https://finance.yahoo.com/> (accessed on 21 January 2026).

**Conflicts of Interest:** The authors declare no conflicts of interest.

## Appendix A. Model Parameter Counts

Table A1 reports the number of trainable parameters for each model variant. XAttnFusion is the largest model due to the cross-attention projection matrices. The ConcatBaseline and single-branch models are substantially smaller, confirming that the performance gains in Section 7 are not simply a consequence of model capacity.

**Table A1.** Trainable parameter counts by model variant.

| Model          | Parameters |
|----------------|------------|
| XAttnFusion    | 692,225    |
| NoCrossAttn    | 487,745    |
| ConcatBaseline | 172,193    |
| MarketOnly     | 123,713    |
| StructureOnly  | 89,217     |
| MacroOnly      | 86,401     |

## Appendix B. Fair Ablation: Per-Fold AUROC

Table A2 reports the per-fold AUROC for each model under its best independently tuned configuration (Section 7). This provides visibility into fold-level variance and shows that XAttnFusion’s advantage is not driven by a single fold.

**Table A2.** Per-fold AUROC under fair ablation (each model at its best configuration).

| Model          | Fold 1 | Fold 2 | Fold 3 | Fold 4 | Fold 5 | Fold 6 |
|----------------|--------|--------|--------|--------|--------|--------|
| XAttnFusion    | 0.639  | 0.680  | 0.597  | 0.705  | 0.568  | 0.611  |
| ConcatBaseline | 0.624  | 0.580  | 0.453  | 0.702  | 0.565  | 0.593  |
| NoCrossAttn    | 0.613  | 0.505  | 0.638  | 0.663  | 0.557  | 0.553  |
| MacroOnly      | 0.566  | 0.586  | 0.595  | 0.510  | 0.554  | 0.496  |
| MarketOnly     | 0.572  | 0.585  | 0.578  | 0.598  | 0.500  | 0.566  |
| StructureOnly  | 0.504  | 0.475  | 0.477  | 0.541  | 0.553  | 0.495  |

Notes: Values rounded to three decimal places. Each model is tuned over 48 hyperparameter configurations (dropout  $\times$  weight decay  $\times$  gradient clipping). The best configuration per model is selected by mean AUROC across folds.

## Appendix C. Fair Ablation: Best Hyperparameter Configurations

Table A3 reports the hyperparameter configuration selected for each model during the fair ablation study. All models were searched over the same grid: dropout  $\in \{0.05, 0.10, 0.15, 0.20\}$ , weight decay  $\in \{10^{-5}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$ , gradient clipping  $\in \{0.5, 1.0, 2.0\}$ . Fixed parameters shared across all models: batch size = 256, learning rate =  $5 \times 10^{-4}$ , AdamW optimizer, ReduceLROnPlateau scheduler (factor 0.5, patience 3), max epochs = 60, early stopping patience = 8, loss = BCE +  $0.5 \times$  Soft-AUC.

**Table A3.** Best hyperparameter configuration per model (fair ablation).

| Model          | Dropout | Weight Decay       | Grad Clip |
|----------------|---------|--------------------|-----------|
| XAttnFusion    | 0.10    | $5 \times 10^{-4}$ | 2.0       |
| ConcatBaseline | 0.05    | $10^{-4}$          | 0.5       |
| NoCrossAttn    | 0.15    | $10^{-3}$          | 1.0       |
| MacroOnly      | 0.05    | $10^{-3}$          | 0.5       |
| MarketOnly     | 0.15    | $10^{-5}$          | 1.0       |
| StructureOnly  | 0.15    | $5 \times 10^{-4}$ | 2.0       |

Notes: No single regularization pattern dominates: XAttnFusion selects moderate dropout (0.10) with higher weight decay ( $5 \times 10^{-4}$ ), while ConcatBaseline prefers low dropout (0.05) with lower weight decay ( $10^{-4}$ ). The diversity of selected configurations across models supports the fair ablation design of independent tuning per model.

## Appendix D. Structure Feature Ablation: Per-Fold Detail

Table A4 reports the per-fold comparison between XAttnFusion with and without the structure branch, evaluated by zeroing the structure inputs at inference time on the tuned checkpoints. Structure features provide a positive AUROC lift in 5 of 6 folds.



**Table A4.** Structure feature ablation (per-fold AUROC).

| Fold | Full Model | No Structure | Lift    |
|------|------------|--------------|---------|
| 1    | 0.6390     | 0.6298       | +0.0091 |
| 2    | 0.6797     | 0.6609       | +0.0188 |
| 3    | 0.5972     | 0.5931       | +0.0042 |
| 4    | 0.7049     | 0.7165       | −0.0116 |
| 5    | 0.5676     | 0.5439       | +0.0238 |
| 6    | 0.6114     | 0.5716       | +0.0398 |
| Mean | 0.6333     | 0.6193       | +0.0140 |

Notes: The full model uses the best tuned configuration (dropout = 0.1, weight decay =  $5 \times 10^{-4}$ , grad clip = 2.0). The “No Structure” column zeroes the structure branch inputs at inference time, keeping all other weights fixed. The lift is positive in 5 of 6 folds.

## Appendix E. Pseudocode for Macro–Market Cross-Attention

The following pseudocode describes the forward pass of XAttnFusion for a single sample at time  $t$ . For clarity, we show single-head attention; the implementation uses  $H = 4$  heads with concatenation and output projection.

```

Input: X_mkt  (82 x 40)  -- market features over 40-day window
      X_mac  (24 x 13)  -- macro features over 24-month window
      X_str  (24,)      -- structure features (scalar vector)

# 1. Market branch (1D-CNN)
H = Conv1d(X_mkt.T, channels=[64, 128], kernel=3) # (128 x 40)
m = [AvgPool(H); MaxPool(H)]                     # (256,)
m = Linear(m)                                     # (256,)

# 2. Macro branch (TCN, 3 blocks, dilation 1,2,4)
Z = TCN(X_mac.T, channels=128, blocks=3)          # (128 x 24)
H_mac = Conv1x1(Z).T                             # (24 x 256)

# 3. Structure branch (MLP)
s = ReLU(Linear(X_str, 64))                       # (64,)
s = Linear(s)                                     # (256,)

# 4. Cross-attention fusion
q = m + s                                         # (1 x 256)
K = H_mac @ W_K                                 # (24 x 64)
V = H_mac @ W_V                                 # (24 x 64)
q_proj = q @ W_Q                                # (1 x 64)

alpha = softmax(q_proj @ K.T / sqrt(64))         # (1 x 24)
c = alpha @ V                                   # (1 x 64)
c = Concat(c_head1, ..., c_head4) @ W_O         # (256,)

# 5. Classifier
h = ReLU(Linear(LayerNorm(c), 64))              # (64,)
logit = Linear(h, 1)                            # scalar

Output: logit, alpha

```

## Appendix F. Macro Variable Descriptions

Table A5 lists the 13 macroeconomic variables used in the macro branch, their FRED series identifiers (where applicable), and their transformations.

**Table A5.** Macroeconomic variables.

| Variable             | FRED ID      | Frequency | Transformation    |
|----------------------|--------------|-----------|-------------------|
| Federal Funds Rate   | FEDFUNDS     | Monthly   | Level             |
| Yield Slope (10Y–2Y) | DGS10, DGS2  | Daily     | Spread            |
| CPI YoY              | CPIAUCSL     | Monthly   | 12-month % change |
| Core CPI YoY         | CPILFESL     | Monthly   | 12-month % change |
| PCE YoY              | PCEPI        | Monthly   | 12-month % change |
| Core PCE YoY         | PCEPILFE     | Monthly   | 12-month % change |
| NFP Monthly Change   | PAYEMS       | Monthly   | First difference  |
| NFP YoY Change       | PAYEMS       | Monthly   | 12-month % change |
| Unemployment Rate    | UNRATE       | Monthly   | Level             |
| Participation Rate   | CIVPART      | Monthly   | Level             |
| High Yield OAS       | BAMLH0A0HYM2 | Daily     | Level             |
| Financial Conditions | NFCI         | Weekly    | Level             |
| WTI Crude Oil        | DCOILWTICO   | Daily     | Level             |

Notes: All series are sourced from the Federal Reserve Economic Data (FRED) API. Daily and weekly series are forward-filled to daily frequency; all series are then resampled to monthly (last value per calendar month) to form the 24-month macro input sequence. To prevent look-ahead bias from publication delays, the macro window is lagged by two months (Section 4.2). Standardization is performed per fold using training-set statistics.

## Appendix G. Data Sources and Reproducibility

Table A6 summarizes all external data sources. Table A7 reports the full configuration used to generate the feature store and train models.

**Table A6.** Data sources.

| Data Type                              | Source        | Period    |
|----------------------------------------|---------------|-----------|
| Market (SPY, QQQ, IWM, TLT, HYG, VIXY) | Yahoo Finance | 2008–2024 |
| Macroeconomic (13 series)              | FRED API      | 2004–2024 |

**Table A7.** Reproducibility configuration.

| Parameter           | Value                               |
|---------------------|-------------------------------------|
| Random seed         | 1008                                |
| Market window       | 40 trading days                     |
| Macro window        | 24 months                           |
| Forecast horizon    | 5 trading days                      |
| Minimum move filter | 1%                                  |
| Macro lag           | 2 months                            |
| CV folds            | 6                                   |
| Embargo             | 5 trading days                      |
| Feature selection   | Top-70 MI + 14 essential (per fold) |
| Embedding dimension | 256                                 |
| Attention heads     | 4                                   |
| Market CNN channels | [64, 128]                           |
| TCN channels        | [128, 128, 128]                     |
| Kernel size         | 3                                   |
| Batch size          | 256                                 |
| Learning rate       | $5 \times 10^{-4}$                  |
| Optimizer           | AdamW                               |

Table A7. Cont.

| Parameter               | Value                                                                             |
|-------------------------|-----------------------------------------------------------------------------------|
| Weight decay            | $5 \times 10^{-4}$ (best); grid $\{10^{-5}, 10^{-4}, 5 \times 10^{-4}, 10^{-3}\}$ |
| Gradient clipping       | 2.0 (best); grid $\{0.5, 1.0, 2.0\}$                                              |
| Loss                    | BCE + $0.5 \times$ Soft-AUC                                                       |
| Max epochs              | 60                                                                                |
| Early stopping patience | 8                                                                                 |
| LR scheduler            | ReduceLROnPlateau (factor 0.5, patience 3)                                        |

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